Evaluating the Quality of Explanations under a Cognitive Feature Budget

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ABSTRACT

Local explanation methods are often evaluated using complete additive reconstructions, although user-facing explanations typically display only a small subset of features. This creates a gap between full fidelity and displayed fidelity. We introduce Complexity-Calibrated Local Concordance (CCC), a budget-aware metric that measures how accurately the top-K additive contributions reconstruct the black-box prediction. CCC evaluates compact faithfulness: the fidelity of the explanation after imposing a cognitive feature budget $K \in \{5, 6, 7, 8, 9\}$. We implement CCC together with standard comparison metrics and apply it to LIME, SHAP, and MAPLE on UCI Breast Cancer Wisconsin Diagnostic and Adult datasets. The experiments show that full fidelity does not guarantee faithful compact explanations, and that explainers differ substantially in how quickly their top-ranked features recover the model prediction. CCC therefore complements existing XAI metrics by making the trade-off between fidelity and human-sized explanations explicit.

KEYWORDS

Explainable AI, local explanations, faithfulness, cognitive complexity

1 INTRODUCTION

Explainable Artificial Intelligence (XAI) is used to make black-box predictions inspectable by users, developers, and domain experts. In tabular classification, local explanation methods such as LIME [10], SHAP [6], and MAPLE [9] are particularly common because they express each prediction as a set of feature contributions. The central evaluation problem is whether those contributions are faithful to the model rather than merely plausible to a human observer [4].

The EQE project (Evaluating the Quality of Explanations) focuses on a specific blind spot in this evaluation problem. Many faithfulness metrics evaluate the complete explanation, but complete explanations can contain all available features. A user-facing explanation, instead, should remain useful after being reduced to a small number of high-impact features. This is not only a design preference: Miller's classic work on working-memory limits suggests that humans can actively process only a small number of information units at once [7]. For this reason, the project evaluates explanations under a feature budget $K \in \{5, 6, 7, 8, 9\}$, aligning with Miller's cognitive limit range.

The contribution is a benchmark and metric centered on *compact faithfulness*. We introduce Complexity-Calibrated Local Concordance (CCC), which measures how well a top-K truncated additive explanation reconstructs the black-box probability. The project implements CCC together with standard comparison metrics and applies it to LIME, SHAP, and MAPLE on the UCI Breast Cancer Wisconsin Diagnostic and Adult datasets.

2 RELATED WORK

XAI evaluation is usually divided into human, application, and functional levels [4]. Human-grounded evaluation studies whether people can use explanations in simplified tasks; application-grounded evaluation studies utility in a real domain workflow; and functionally-grounded evaluation uses computational proxies when human studies are not available. This project belongs to the third category: it evaluates formal properties of explanations produced for trained classifiers.

2.1 Local Explanation Methods

LIME fits a sparse local surrogate around the instance to be explained [10]. It is model-agnostic and intuitive, but its output can depend strongly on perturbation sampling, feature scaling, and the local linear fit. For additive fidelity evaluation, LIME coefficients are multiplied by the interpretable representation $\mathbf{z}(x)$ to obtain instance-level contributions. SHAP computes feature attributions based on Shapley values and satisfies desirable axioms for additive explanations [6]. In this project SHAP is implemented through KernelExplainer, using a sampled training background and the positive-class probability as the explained output. MAPLE learns local linear explanations by combining random-forest neighborhoods with supervised local regression [9]. In the codebase, MAPLE coefficients are converted into per-instance additive contributions by multiplying each coefficient by the corresponding feature value, so all three explainers share the same interface: weights and intercepts.

2.2 Evaluation Metrics

The literature review conducted for the project identified several families of functionally-grounded metrics. Fidelity metrics compare the explanation with the black-box model. Sufficiency and comprehensiveness perturb relevant features to check whether selected features preserve or remove prediction evidence. Robustness metrics, including sensitivity and infidelity [13], measure how explanations react to input perturbations. Benchmark projects such as XAI-Bench [5] and OpenXAI [1] systematize these criteria across methods and datasets. Frameworks such as LEAF [2] also include local concordance and conciseness criteria. However, these metrics usually treat faithfulness and compactness as separate properties: one metric asks whether the full explanation is faithful, another asks whether the explanation is small. CCC combines the two by asking whether the explanation is still faithful *after* enforcing a small top-K budget.

3 RESEARCH GAPS

The project targets the gap between mathematical fidelity and usable fidelity. For an additive explanation $g(x) = w_0 + \sum_i \phi_i(x)$, a method may be perfectly faithful when every feature contribution is included. SHAP, for example, is designed to satisfy local accuracy under the full additive reconstruction. But a practitioner may show only the largest five to nine contributions in a dashboard, report, or interactive tool. If the omitted contributions collectively matter, the compact explanation can misrepresent the model even though the full explanation is exact.

This gap is not captured by model accuracy either. A high-accuracy classifier can be difficult to explain compactly, while a slightly less accurate model may produce local attributions that concentrate prediction evidence in fewer features. The research question is therefore:

> Given a black-box model and a local additive explainer, how much fidelity is preserved when only the top-K feature contributions are shown?

Answering this question helps compare explainers under a realistic cognitive budget and provides a sanity check for feature rankings.

4 METHODOLOGY

4.1 Complexity-Calibrated Local Concordance

For each test instance x, the framework stores the black-box positive-class probability $f(x)$, the explainer intercept w_0, and the additive contribution vector $\phi(x)$. Let $S_K(x)$ be the indices of the K largest absolute contributions. The truncated local reconstruction is

$$g_K(x) = w_0 + \sum_{i \in S_K(x)} \phi_i(x). \tag{1}$$

Complexity-Calibrated Local Concordance is the mean squared error between the black-box output and this truncated reconstruction:

$$\mathrm{CCC}_K = \frac{1}{N} \sum_{j=1}^{N} \left(f(x^{(j)}) - g_K(x^{(j)}) \right)^2. \tag{2}$$

Lower CCC means that the explainer preserves local predictive fidelity using only K features.

The implementation intentionally does not multiply contributions by the raw features during reconstruction: SHAP values are already additive contributions, LIME returns local feature weights that are multiplied by the interpretable representation $\mathbf{z}(x)$ to obtain instance-level additive contributions $\phi_i = \beta_i \cdot z_i(x)$, and MAPLE coefficients are converted into contributions inside the explainer wrapper. This normalization is essential because CCC assumes a common additive form across methods.

4.2 Explainer Conversion Protocol

The framework normalizes different explainers into a common additive form $g(x) = w_0 + \sum_i \phi_i(x)$. For SHAP, values are already additive contributions. For LIME, the surrogate model coefficients are multiplied by the interpretable representation $\mathbf{z}(x)$ to obtain instance-level contributions $\phi_i = \beta_i \cdot z_i(x)$. For MAPLE, each coefficient β_i is multiplied by the corresponding feature value to obtain $\phi_i = \beta_i \cdot x_i$. All three return weights and intercepts in the same format expected by CCC. Table 1 summarizes the implemented metrics. Full MSE measures the fidelity of the complete additive explanation. Random-K MSE keeps K random contributions from the same weight vector and is repeated 30 times; it tests whether the explainer's top-K ordering is better than arbitrary selection.

Table 1: Metrics implemented in `core/metrics.py`.

Metric	Quantity measured	Direction
CCC	Top-K reconstruction error	lower
Full MSE	Full additive reconstruction error	lower
Random-K	Random top-K baseline	lower
Sufficiency	Drift when only top-K inputs remain	lower
Comprehensive.	Drift when top-K inputs are removed	higher
Normalized CCC	CCC divided by prediction variance	lower
Degradation	CCC divided by full MSE	lower

Sufficiency and comprehensiveness use a feature baseline, implemented as the mean training vector, to evaluate prediction drift under input masking. The current codebase also includes normalized CCC (CCC divided by the variance of black-box probabilities, enabling cross-dataset comparison) and top-K degradation ratio for cross-dataset comparison and truncation analysis.

5 EXPERIMENTS AND ANALYSIS

5.1 Experimental Setup

The main benchmark uses two public datasets: UCI Breast Cancer Wisconsin Diagnostic [11] and UCI Adult [12]. The classifiers are XGBoost and a neural network with two hidden layers of 100 units and a maximum of 500 training iterations. The explainers are LIME, SHAP KernelExplainer, and MAPLE. SHAP uses a sampled training background of size 100. MAPLE uses random forest feature extraction, 100 estimators, a validation size of 100, a training subsample of 2000 when available, and regularization 0.001. The benchmark evaluates $K = 5, 6, 7, 8, 9$ over seeds 42, 123, and 2026, focusing on Miller's cognitive limit range.

The results discussed below come from `results/latest.md`, which contains aggregate tables grouped by dataset, model, explainer, and K. The run reports six dataset–seed executions and includes model accuracy plus CCC, full MSE, sufficiency, comprehensiveness, and random-K MSE.

5.2 Software Architecture

The project is organized as a reproducible Python benchmark. The entry point `core/main.py` reads `config.yml`, builds a matrix of dataset–seed runs, executes them in parallel with joblib, and writes Markdown results. The experiment orchestration is implemented in `core/test_framework.py`. For each dataset and seed, the orchestrator: loads and scales the data, trains every configured model, generates explanations for every model/explainer pair, caches the weights and intercepts, and computes every metric for all configured values of K.

The data loaders support the local copies of the UCI Breast Cancer Wisconsin Diagnostic and Adult datasets. Breast Cancer uses the WDBC files and recovers descriptive feature names from metadata; Adult combines the train and test files, removes missing values, one-hot encodes categorical attributes, and standardizes features. The model wrappers expose a uniform `train`, `predict`, and

Table 2: Mean model accuracy across the three seeds.

Dataset	Model	Accuracy
Adult	XGBoost	0.874
Adult	Neural network	0.823
Breast Cancer	XGBoost	0.968
Breast Cancer	Neural network	0.977

`predict_proba` interface for XGBoost [3] and a scikit-learn multilayer perceptron [8]. The visualization module generates summary charts from result files and per-instance top-K reduction figures for presentation.

Model accuracy does not inform explainer selection. A high-accuracy classifier can be difficult to explain compactly, while a slightly less accurate model may produce local attributions that concentrate prediction evidence in fewer features. On Breast Cancer, SHAP explanations for XGBoost reach lower compact error (CCC) at small and large K. On Adult, XGBoost is more accurate than the neural network, but CCC still varies primarily with explainer choice rather than accuracy.

Figure 1 and Table 3 show three main patterns. First, SHAP achieves the lowest CCC scores in the tested configurations. Its error decreases rapidly as K increases, reaching near-zero values at $K = 9$ on Adult/XGBoost and Breast Cancer/XGBoost. This is expected from SHAP's additive local accuracy, but the non-zero values at small K are the important observation: a full SHAP explanation can be exact while its compact version still loses fidelity.

Second, MAPLE is an intermediate baseline, with behavior that depends on the dataset. On Breast Cancer/XGBoost, increasing K reduces CCC from 0.0755 to 0.0341, meaning that the top-ranked contributions recover some predictive signal. On Adult/NN and Adult/XGBoost, however, MAPLE is almost flat across K. This suggests that the ranking does not concentrate the reconstruction signal in a small subset of features, even if the full local approximation is not terrible.

Third, LIME is unstable across settings. On Adult/NN, LIME CCC remains above 3.0 even at $K = 9$ (value 2.957), several orders of magnitude worse than SHAP. This indicates that the surrogate model coefficients do not reconstruct the black-box probability well in the additive form $g_K(x) = w_0 + \sum_{i \in S_K} \phi_i$. On Breast Cancer/XGBoost, LIME is closer to SHAP for very small budgets but does not improve monotonically with K: the error decreases until about $K = 5$ or $K = 6$ and then rises again. This behavior indicates that adding more locally weighted features does not necessarily make the truncated reconstruction closer to the black-box probability.

5.3 Top-K versus Random-K

The random-K baseline gives CCC an internal sanity check. In most tested settings, the explainer-selected top-K features outperform random subsets of the same explanation weights. For example, Adult/XGBoost SHAP at $K = 9$ has CCC around 3.2×10^{-5}, while random-K MSE is about 0.086. Breast Cancer/XGBoost SHAP at $K = 9$ has CCC around 0.00028, while random-K is about 0.113. This confirms that the top-ranked features preserve meaningful prediction signal. When the gap is smaller, as for some MAPLE configurations, the feature ranking is less informative.

5.4 What CCC Reveals

CCC shifts the evaluation target from "Is the full explanation faithful?" to "Is the displayed explanation faithful?" The main insight is that CCC separates three notions that can otherwise be confused: model accuracy, full local fidelity, and compact local fidelity. SHAP's full MSE is approximately zero because the full additive reconstruction matches the explained model output, but CCC still exposes the cost of showing only five to nine features. MAPLE can have moderate full MSE but poor top-K concentration. LIME can produce local weights whose selected features do not reconstruct the probability well. These distinctions are directly relevant for systems that display explanations to humans, because those systems normally display only a subset of features.

5.5 Does CCC add useful information?

Although full fidelity indicates whether the complete additive explanation reconstructs the model output, it does not evaluate the quality of the explanation after top-K truncation. CCC adds this missing information by measuring how much predictive fidelity is preserved when only a cognitively limited number of features is shown. CCC differs from existing metrics in two ways. First, it operates on additive explanations and measures reconstruction error directly, similar to LEAF's local concordance but with an explicit complexity budget. Second, unlike infidelity or AOPC/deletion metrics that perturb inputs, CCC evaluates whether the top-K feature contributions alone reconstruct the black-box probability. Two explanations with identical full MSE can have different CCC values, revealing distinct levels of compact faithfulness.

Comparison with full MSE. Table 5 shows aggregate values where full MSE is invariant across K while CCC changes. For example, on Breast Cancer/XGBoost, SHAP has full MSE of zero for all K, but CCC decreases from 0.0160 at $K = 5$ to 0.0003 at $K = 9$. This demonstrates that a perfect full-explanation fidelity does not guarantee faithful compact explanations.

Overall, CCC complements existing XAI metrics by evaluating a property that the others do not capture: compact faithfulness under a fixed feature budget. While full MSE measures complete explanation fidelity and sufficiency/comprehensiveness measure feature perturbation effects, CCC specifically asks whether the truncated additive explanation remains faithful to the black-box model. This makes CCC particularly relevant for practical XAI settings where explanations must be both faithful to the model and short enough to be inspected by a human.

CCC can also be interpreted as a family of metrics parameterized by the feature budget. The full CCC curve $\{CCC_K\}_{K=5}^{9}$ describes how reconstruction error decreases as more features are included. This curve can be summarized by the AUC-CCC (area under the curve) or by the minimum faithful K needed to reach a model-specific error threshold, enabling CCC to serve as a runtime quality-control signal for deployed explanation tools.

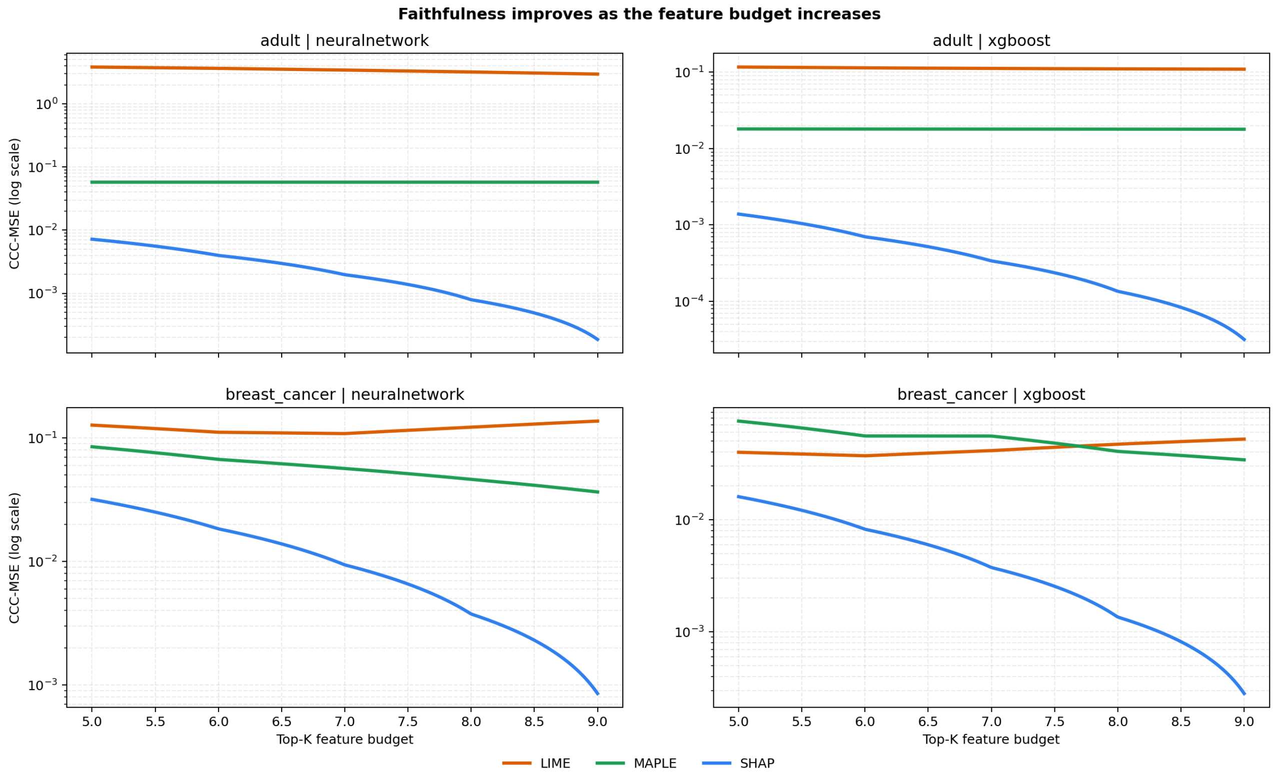

Figure 1: CCC as the feature budget increases. The figure was generated from the project visualization utilities. Lower curves indicate more faithful compact explanations.

Table 3: Representative aggregate CCC values from `results/latest.md`. Lower is better.

Dataset/model	Explainer	$K = 5$	$K = 6$	$K = 7$	$K = 8$	$K = 9$
Adult/NN	SHAP	0.007182	0.003973	0.001968	0.000790	0.000186
Adult/NN	MAPLE	0.0574	0.0574	0.0574	0.0574	0.0574
Adult/NN	LIME	3.8262	3.6462	3.4261	3.1941	2.9571
Adult/XGB	SHAP	0.001387	0.000699	0.000338	0.000135	0.000032
Adult/XGB	MAPLE	0.0181	0.0180	0.0179	0.0179	0.0179
Adult/XGB	LIME	0.1168	0.1139	0.1119	0.1104	0.1095
Breast/NN	SHAP	0.0318	0.0183	0.0094	0.0037	0.0009
Breast/NN	MAPLE	0.0847	0.0670	0.0565	0.0462	0.0365
Breast/NN	LIME	0.1269	0.1111	0.1084	0.1220	0.1368
Breast/XGB	SHAP	0.0160	0.0082	0.0037	0.0014	0.0003
Breast/XGB	MAPLE	0.0755	0.0556	0.0555	0.0405	0.0341
Breast/XGB	LIME	0.0398	0.0370	0.0412	0.0470	0.0521

5.6 Limitations

CCC applies naturally to additive explanations, but it is not directly suitable for rule lists, counterfactual sets, prototypes, or explanations whose output is not a decomposable sum. The metric also depends on the scale and calibration of the model probability. For this reason the code includes normalized CCC, which divides the error by the variance of black-box probabilities, although the main archived run discussed here reports raw CCC. Perturbation metrics such as sufficiency and comprehensiveness depend on the masking baseline; the current implementation uses the mean training vector, which is simple and reproducible but may create unrealistic inputs for correlated features. The feature budget K is an operational proxy for cognitive complexity following Miller's "7 ± 2" range, but showing K one-hot-encoded features does not equal exposing K semantic concepts, which limits interpretation on Adult. Finally, the benchmark is limited to two datasets and two model families. The

Table 4: CCC compared with existing evaluation metrics. While other metrics treat fidelity and compactness as separate properties, CCC evaluates their intersection: fidelity under an explicit cognitive budget.

Metric	Relation to CCC
Full MSE/local accuracy	Evaluates all-feature reconstruction; CCC measures the reconstruction after top-K truncation.
Sufficiency	Keeps top-K input features under a masking baseline; CCC evaluates additive concordance directly.
Comprehensiveness	Measures prediction change after removing features; CCC asks whether displayed additive terms reconstruct $f(x)$.
Infidelity/sensitivity	Measures stability under input perturbations; CCC adds an explicit cognitive feature budget.
Conciseness	Measures explanation size alone; CCC combines compactness with predictive faithfulness.

Table 5: CCC vs full MSE for SHAP on Breast Cancer/XGBoost. Full MSE remains constant while CCC varies with K.

K	5	6	7	8	9
full MSE	0.0	0.0	0.0	0.0	0.0
CCC	0.0160	0.0082	0.0037	0.0014	0.0003

Table 6: CCC vs sufficiency/comprehensiveness on Adult/XGBoost with $K = 8$. Different metrics produce different explainer rankings.

Explainer	CCC	Sufficiency	Comprehensiveness
SHAP	0.000135	0.0624	0.2278
MAPLE	0.0179	**0.0334**	0.2299
LIME	0.1104	0.1015	0.2054

Table 7: CCC vs random-K baseline on Adult/XGBoost with $K = 9$. Top-K selection is much better than random feature subsets.

Explainer	CCC	Random-K MSE
SHAP	0.000032	0.0863
MAPLE	0.0179	0.0385
LIME	0.1095	0.5266

conclusions are therefore strongest as evidence for the proposed evaluation protocol, not as a universal ranking of all XAI methods.

6 CONCLUSIONS

EQE contributes a compact-faithfulness evaluation of local explanations through Complexity-Calibrated Local Concordance. This work makes three contributions. First, it introduces CCC, a budget-conditioned local fidelity metric that measures whether the top-K displayed additive contributions reconstruct the black-box prediction. Second, it shows empirically that full additive fidelity and compact additive fidelity are distinct evaluation targets: a method can achieve near-zero full MSE while still losing fidelity after truncation. Third, it provides a benchmark protocol comparing top-K CCC against random-K baselines, sufficiency, comprehensiveness, and full MSE across datasets, models, and explainers. The benchmark shows that SHAP achieves the lowest CCC scores in the tested configurations, MAPLE is a useful intermediate comparison, and LIME is less reliable in the tested configurations. More importantly, the results show that full-explanation fidelity and compact-explanation fidelity are distinct evaluation targets. This makes CCC particularly relevant for practical XAI settings where explanations must be both faithful to the model and short enough to be inspected by a human.

The most immediate extension is to broaden the benchmark without changing the metric definition. Additional tabular datasets would test whether CCC remains stable when the number of features, categorical sparsity, and class imbalance change. Additional black-box models, such as calibrated linear models, random forests, and gradient-boosted variants, would also clarify whether compact faithfulness depends more on the explainer or on the geometry of the learned decision function.

A second direction is to make the feature budget adaptive. The current report uses fixed values from five to nine because they are easy to compare and align with a cognitive-complexity motivation. In an interface, however, a useful system could choose the smallest K whose CCC falls below a model-specific threshold, or expose a warning when no compact explanation reaches acceptable fidelity. This would turn CCC from an offline benchmark metric into a runtime quality-control signal for deployed explanation tools.

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